Unit Test Report

Automated pytest suite | 6 November 2026

1. Summary

211 tests run, 211 passed, 0 failed. Environment: Windows 11, Python 3.14, pytest 9 with pytest-cov, run on 6 November 2026 against the main branch. The same suite runs on every pull request in GitHub Actions on Ubuntu, Windows and macOS with Python 3.10 and 3.12.

Unit tests call functions directly (for example the fee rule and BFS). Integration tests drive the Flask app through its test client with a fresh temporary SQLite database for every test, so tests never depend on each other.

2. Results by test file

Test file	What it covers	Tests	Passed	Failed	Time (s)
test_setup.py	App start, tables, seed layout, accounts, persistence	5	5	0	1.9
test_grid.py	BFS paths, ties, walls, grid size, timing	14	14	0	1.6
test_entry.py	Plate validation, ticket ID, QR, duplicate entry	20	20	0	8.0
test_exit_fee.py	Fee rules, grace boundaries, exit API	18	18	0	3.7
test_slots.py	Slot categories, live grid data, reservations	12	12	0	5.0
test_auth.py	Login, logout, roles, redirects	14	14	0	6.7
test_payment.py	Mock payment, forced failure, receipt	9	9	0	4.0
test_demo.py	Seed data, Demo Clock, overstay	11	11	0	6.4
test_route.py	Route preview, exit mode, reassignment	9	9	0	3.5
test_performance.py	Parallel entries, double payment, timing	6	6	0	2.4
test_reports.py	Reports, CSV and PDF export, validation	17	17	0	10.2
test_prediction.py	Wait estimate formula and messages	10	10	0	4.1
test_admin.py	Settings, layout editor, audit log	22	22	0	12.7
test_security.py	Lockout, sessions, CSRF, injection, escaping	18	18	0	13.3
test_plan_cases.py	Test Plan TC-01 to TC-17 (system level)	26	26	0	17.1
Total		211	211	0	100.4

3. Techniques used

Equivalence partitioning and boundary values (plates of 3, 4, 10 and 11 characters; stays of 4:59, 5:00 and 5:01; grace 0 and 30), decision tables (fees by vehicle type), state transitions (slot Vacant, Occupied, Reserved; ticket Active, Closed), concurrency with threads, and negative tests for every error code.

4. How to reproduce

python -m pytest -v --junitxml=junit.xml

Wall-clock timing checks are skipped on GitHub Actions and run on the developer machine; see Deviations D-19.